

VERIFICATION SUPPLEMENT FOR EXACT NON-SCATTERING HOMOGENEOUS DIELECTRICS FROM SCHIFFER COUNTEREXAMPLES

MANSEHEJ SINGH

ABSTRACT. This supplement records the computer-assisted and formal-verification layer supporting the accompanying paper. It specifies the imported Colbrook–Stepaniants theorem, derives the conservative preconditioner bound $\|\mathcal{A}\| < 1180$, gives the detailed Bessel perturbation estimates, normalizes the endpoint inequalities by exact rational arithmetic, and documents the independent executable verifiers and the narrow Lean 4 re-verification of the certificate arithmetic. The supplement is deliberately separated from the research article: the article presents the mathematical discovery and proof, while this document provides the audit trail needed to replay and challenge the certificate.

1. LOGICAL STRUCTURE AND SCOPE

The proof has three layers.

- (i) The theorem of Colbrook and Stepaniants [CS26] supplies an exact noncircular Schiffer zero in a weighted coefficient algebra, together with an invertible preconditioner, contraction bounds, infinite-tail control, and uniform geometry estimates.
- (ii) The new analytic layer defines the Bessel-deformed equation $F_\tau = 0$, derives the value and derivative perturbation estimates, and proves that the same contraction ball remains valid for $0 \leq \tau \leq 10^{-13}$.
- (iii) The physical layer pushes the exact fixed point through the conformal map and a dilation, yielding a constant-index transmission solution whose exterior scattered field is identically zero.

The large upstream interval computation is not reimplemented in this supplement. Instead, the released upstream certificate is treated as a pinned theorem dependency and replayed from its frozen source and inverse. The new scalar inequalities are checked independently by exact-rational Python, Node/BigInt arithmetic, and Lean normalization.

2. PINNED UPSTREAM THEOREM

The upstream fixed-disc equation is

$$(1) \quad F_0(\gamma, p) = \gamma + |p|^2 (1 + K\gamma) = 0,$$

with ten-fold shape series

$$p(z) = \sum_{j \geq 0} p_j z^{10j}, \quad \phi_p(z) = z + \sum_{j \geq 1} \frac{p_j}{p_0(10j+1)} z^{10j+1}.$$

The certificate parameters are

$$(L, S, R, J, \rho) = (60, 40, 30, 30, 1.05), \quad N = 2471,$$

and the validated radius is

$$r = 10^{-6}.$$

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The authenticated coarse majorants are

$$(2) \quad Y \leq 1.59 \times 10^{-10}, \quad Z \leq 0.621, \quad C_2 \leq 122, \quad C_3 \leq 0.012.$$

They imply

$$(3) \quad R_0 := Y + Zr + C_2r^2 + C_3r^3 - r \leq -0.00000037871899999998,$$

$$(4) \quad L_0 := Z + 2C_2r + 3C_3r^2 \leq 0.621244000000036.$$

The same theorem proves on the whole validated ball

$$(5) \quad \|p\|_\rho < 55, \quad 31 < p_0 < 32, \quad \sum_{j \geq 1} |p_j/p_0| < 0.65,$$

and keeps the first nonconstant shape coefficient away from zero. Hence $\operatorname{Re} \phi'_p > 0.35$ on the closed unit disc, so ϕ_p is univalent on a complex collar and its image is a noncircular analytic Jordan domain.

The release pins the following upstream objects:

- repository and commit;
- exact hexadecimal center;
- frozen binary64 approximate inverse;
- source manifest and certificate schema;
- authenticated 192-bit and 256-bit MPFR audit hashes.

The concrete values are stored in `verification/upstream.lock` and `verification/upstream_snapshot.json`.

3. HARDENING THE PRECONDITIONER BOUND

Let

$$b = \bar{p}_0, \quad \alpha = \max(1, \|\mathcal{A}\|).$$

The upstream nonlinear estimate defines

$$(6) \quad C_3 = \frac{\alpha}{96b^2}.$$

The exact first shape token is

$$b = 0x1.ff78dc9f663aap+4 = \frac{4498956314423765}{140737488355328} < 32.$$

Combining this exact inequality with the simple authenticated majorant $C_3 \leq 0.012$ yields

$$(7) \quad \begin{aligned} \alpha &= 96b^2C_3 \\ &< 96 \cdot 32^2 \cdot 0.012 = 1179.648 < 1180. \end{aligned}$$

Thus the extension proof needs neither the full inverse norm nor a long decimal reconstructed from the explanatory paper. The executable certificate uses the integer upper bound

$$(8) \quad \boxed{\|\mathcal{A}\| < 1180.}$$

4. DETAILED BESSEL PERTURBATION ESTIMATES

For $\tau \geq 0$, set

$$(9) \quad H_{\tau,p} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{\tau |\phi_p|^2}{4} \right)^n$$

and

$$(10) \quad F_{\tau}(\gamma, p) = \gamma + |p|^2 K \gamma + |p|^2 \left(1 - \frac{\tau}{p_0^2} \right) H_{\tau,p}.$$

Write $\delta F_{\tau} = F_{\tau} - F_0$.

4.1. Bound on the conformal radius factor. Write $\phi_p(z) = z\psi_p(z)$. From the coefficient formula and (5),

$$(11) \quad \begin{aligned} \|\psi_p\| &\leq 1 + \sum_{j \geq 1} \frac{|p_j|}{p_0(10j+1)} \rho^j \\ &\leq 1 + \frac{\|p\|_{\rho} - p_0}{11p_0} < 1 + \frac{55 - 31}{11 \cdot 31} < \frac{11}{10}. \end{aligned}$$

The upstream algebra norm is a weighted ℓ^1 sum over a basis indexed by an angular block index ℓ and a radial index s , with weight $c_{\ell}\rho^{\ell}$ depending only on ℓ , and multiplication by the radial factor $|z|^2 = r^2$ acts as the pure radial shift $(\ell, s) \mapsto (\ell, s+1)$ [CS26c, Validation Specification, §§2 and 4]. Multiplication by $|z|^2$ is therefore nonexpansive, and

$$(12) \quad \left\| |\phi_p|^2 \right\| = \left\| |z|^2 |\psi_p|^2 \right\| \leq \|\psi_p\|^2 < \frac{121}{100} < \frac{5}{4}.$$

This radial-shift property is the single algebra fact imported beyond submultiplicativity; it is restated as a lemma in the main article and is item one of the recommended independent audits.

4.2. Value perturbation. Put

$$u = \frac{\tau}{4} \left\| |\phi_p|^2 \right\|.$$

For $0 \leq \tau \leq 10^{-13}$, $u < 1/2$. Since $1/(n!)^2 \leq 1$, the full Bessel series satisfies

$$(13) \quad \|H_{\tau,p}\| \leq \sum_{n \geq 0} u^n = \frac{1}{1-u} \leq 2.$$

Similarly,

$$(14) \quad \|H_{\tau,p} - 1\| \leq \frac{u}{1-u} \leq 2u \leq \frac{5}{8}\tau.$$

Hence

$$(15) \quad \begin{aligned} \left\| \left(1 - \frac{\tau}{p_0^2} \right) H_{\tau,p} - 1 \right\| &\leq \|H_{\tau,p} - 1\| + \frac{\tau}{p_0^2} \|H_{\tau,p}\| \\ &\leq \left(\frac{5}{8} + \frac{2}{31^2} \right) \tau < \frac{63}{100} \tau. \end{aligned}$$

Using $\|p\|^2 < 55^2 = 3025$,

$$(16) \quad \|\delta F_{\tau}\| < 3025 \cdot \frac{63}{100} \tau = 1905.75\tau < 1906\tau.$$

4.3. Derivative perturbation. Let ξ be a shape variation with full X_ρ norm at most one. The upstream product norm weights the shape block by $2\bar{p}_0\rho^j$ [CS26c, Validation Specification, Eq. (13)], which gives

$$(17) \quad \|\xi\| \leq \frac{1}{2b} < \frac{1}{62}, \quad |\xi_0| < \frac{1}{62}.$$

Differentiating ψ_p gives

$$(18) \quad \begin{aligned} \|D\psi_p[\xi]\| &\leq \frac{1}{11} \left(\frac{\|\xi\|}{p_0} + \frac{\|p\| |\xi_0|}{p_0^2} \right) \\ &< \frac{1}{11} \left(\frac{1}{62 \cdot 31} + \frac{55}{62 \cdot 31^2} \right) < \frac{1}{7000}. \end{aligned}$$

Thus

$$(19) \quad \left\| D|\phi_p|^2[\xi] \right\| \leq 2 \|\psi_p\| \|D\psi_p[\xi]\| < \frac{1}{3000},$$

using the radial-shift property once more for the factor $|z|^2$. Termwise differentiation of (9) and $\sum_{n \geq 1} nu^{n-1} = (1-u)^{-2} \leq 4$ give

$$(20) \quad \|DH_{\tau,p}[\xi]\| \leq \frac{\tau}{4} \left\| D|\phi_p|^2[\xi] \right\| \sum_{n \geq 1} nu^{n-1} < \frac{\tau}{3000}.$$

Let

$$B_{\tau,p} = \left(1 - \frac{\tau}{p_0^2} \right) H_{\tau,p} - 1.$$

Then

$$\begin{aligned} D\delta F_\tau[\xi] &= 2 \operatorname{Re}(\bar{p}\xi) B_{\tau,p} \\ &\quad + |p|^2 \left[\left(1 - \frac{\tau}{p_0^2} \right) DH_{\tau,p}[\xi] + \frac{2\tau\xi_0}{p_0^3} H_{\tau,p} \right]. \end{aligned}$$

Applying (15), (13), (17), and (20),

$$(21) \quad \begin{aligned} \|D\delta F_\tau\| &< \left[\frac{110}{62} \frac{63}{100} + \frac{3025}{3000} + \frac{12100}{62 \cdot 31^3} \right] \tau \\ &< 3\tau. \end{aligned}$$

The last strict inequality is checked exactly in all three arithmetic implementations.

5. EXACT CLOSURE OF THE CONTRACTION BALL

Let

$$T_\tau = I - \mathcal{A}F_\tau.$$

The value perturbation changes the invariant-ball radii polynomial by at most

$$1180 \cdot 1906 \tau,$$

and the derivative perturbation changes the contraction constant by at most

$$1180 \cdot 3 \tau.$$

Thus

$$(22) \quad R_\tau \leq R_0 + 1180 \cdot 1906 \tau,$$

$$(23) \quad L_\tau \leq L_0 + 1180 \cdot 3 \tau.$$

At $\tau_* = 10^{-13}$, exact rational arithmetic yields

$$(24) \quad R_{\tau_*} \leq -\frac{153810999999988}{10^{21}} = -1.53810999999988 \times 10^{-7} < 0,$$

$$(25) \quad L_{\tau_*} \leq \frac{621244000354036}{10^{15}} = 0.621244000354036 < 1.$$

Because the right-hand sides are monotone in τ , the same conclusions hold for the whole interval $0 \leq \tau \leq 10^{-13}$.

For comparison, the explicit example $\tau = 10^{-14}$ satisfies

$$R_\tau \leq -3.56228199999988 \times 10^{-7} < 0,$$

$$L_\tau \leq 0.621244000035436 < 1.$$

6. EXECUTABLE CERTIFICATE

The authoritative new input is `verification/extension_certificate.json`. It contains the rational constants, the parameter interval, the exact hexadecimal center coefficient used in (7), and hashes of the pinned upstream snapshot. The primary checker `verification/verify_extension.py`:

- (1) parses all decimal values as exact rational numbers;
- (2) cross-checks the snapshot hashes, repository, and commit against the pinned `upstream.lock` (file-level hash verification is performed by the upstream replay script below);
- (3) parses the hexadecimal binary64 center token exactly and derives $b < 32$, the ball bound $b + r/(2 \cdot 31) < 32$, and $\alpha < 1180$ rather than trusting them as loose text constants;
- (4) rederives the perturbation-constant chain from the certified ball bounds $\|p\|_\rho < 55$ and $p_0 > 31$: the bounds $11/10$ for ψ_p , $5/4$ for $|\phi_p|^2$, $63/100$ for the datum deviation, $1/7000$ and $1/3000$ for the derivatives, and finally $55^2 \cdot 63/100 < 1906$ and the three-term derivative sum below 3;
- (5) verifies the Bessel-series condition at the uniform endpoint $\tau = 10^{-13}$;
- (6) recomputes the radii polynomial, the contraction constant, and both endpoint inequalities, comparing each against the stored exact decimal expansions;
- (7) fails by explicit exceptions and nonzero exit status.

Proof conditions do not use Python’s removable `assert` statement, so normal and optimized execution are byte-identical.

A separately implemented Node program, `verification/verify_extension.mjs`, uses BigInt rational arithmetic, reads the same authoritative certificate, and replays the complete check set above, including the derivation chain and the lock cross-check.

6.1. Upstream replay. `verification/fetch_and_check_upstream.sh` downloads the archived upstream release from Zenodo (record 21765287), verifies the archive digest recorded in `upstream.lock`, and then runs `verification/cross_check_upstream.py`, which

- (1) verifies the pinned SHA-256 hashes of the upstream center, inverse, certificate, and source manifest;
- (2) compares the upstream certificate values $(Y, Z, C_2, C_3, r, \rho, L, S, R, J)$, the univalence sum, and the first-shape-coefficient bound against `upstream_snapshot.json`;
- (3) rederives the ball bounds $31 < p_0 < 32$ and $\|p\|_\rho < 55$ from the exact hexadecimal center coefficients plus the validated radius.

Finally it replays the upstream authenticated checker `verify_certificate.py` itself. The successful run of this entire pipeline from a clean fetch is recorded in `verification/upstream_check.log`; the upstream checker’s exact radii-polynomial output reproduces (3) digit for digit.

6.2. Adversarial tests. The regression suite deliberately corrupts, one at a time: the preconditioner bound; the value-perturbation constant; the conformal-square majorant $5/4$; the endpoint τ ; a stored exact decimal expansion; the upstream Y , Z , and C_3 majorants; the certified shape-norm ball bound 55 ; the exact hexadecimal center token; a pinned hash (forcing a snapshot/lock disagreement); and the presence of the snapshot and lock files themselves. Every corruption must be rejected by the Python checker *and*, when Node is available, by the independent BigInt checker. The suite also requires byte-equal output under `python3` and `python3 -O`.

7. LEAN RE-VERIFICATION OF THE CERTIFICATE ARITHMETIC

The release contains a small Lean 4 project pinned to Lean/mathlib v4.32.1. Its scope is deliberately narrow and is stated here exactly.

Module	Source content
Certificate	Exact rational evaluation of (3)–(25), the hardened $C_3, b \Rightarrow \alpha < 1180$ implication, uniform closure over the whole parameter interval, and the two scalar majorant inequalities $55^2 \cdot 63/100 < 1906$ and the three-term derivative sum below 3.
FixedPoint	Banach fixed-point existence and uniqueness on a complete metric space, and the stability estimate for fixed points of two nearby contractions, stated generically.
LinearRegularity	If A and $A \circ B$ are bijective, so is B .
Scaling	The scalar contrast computation $q = p_0^2/\tau$, its positivity, and $q > 1$ on the certified interval.

What the Lean development is *not*: it does not formalize the operator-level perturbation estimates behind the constants 1906 and 3, the weighted coefficient algebra, the upstream MPFR certificate, or any planar scattering analysis (the Bessel–Herglotz identity, conformal covariance, trace transformation, exterior uniqueness). The generic fixed-point lemmas are stated in the form the paper uses but are not instantiated on the concrete coefficient spaces. The accurate public description is therefore:

A Lean re-verification of the exact rational certificate arithmetic, together with generic fixed-point and invertibility lemmas in the form used by the paper. It is a third independent checker of the scalar arithmetic, not a formalization of the non-scattering theorem.

A static source scan (`formalization/scripts/check_no_holes.py`) rejects `sorry`, `admit`, `axiom`, `native_decide`, `opaque`, `unsafe`, `partial`, and related escape hatches in the project sources, and `NonScattering/MainTheorem.lean` runs `#print axioms` on the public theorems. The packaging environment did not include the Lean toolchain: a fresh pinned-toolchain kernel build, with its log and axiom report archived, remains a pre-publication gate, and until then no claim stronger than “source scan” is made.

8. REPRODUCTION

From the release root, the primary checks are

```
make verify
make lean
make paper
```

The verification-only path is

```
cd verification
./reproduce.sh
```

and the Lean path is

```
cd formalization
python3 scripts/check_no_holes.py
lake build
```

The reproducer authenticates dependencies before executing proof arithmetic. The root manifest covers every regular release file except generated archives and the manifest itself.

9. DEPENDENCY LEDGER

Claim	Status	Evidence
Exact Schiffer zero and infinite tails	Imported theorem	Pinned and replayed Colbrook–Stepanians MPFR certificate.
Univalence, analytic boundary, non-disc	Imported theorem	Uniform geometry bounds on the same certified ball.
Bessel-deformed equation	Article proof	Direct Laplacian and conformal calculation.
Constants 1906 and 3	Article and supplement proof	Derivation chain rechecked in exact rationals by Python and Node; terminal scalar inequalities also in Lean.
Uniform fixed point	Article proof	Banach contraction theorem plus the endpoint certificate; scalar closure rechecked in Python, Node, and Lean.
Scaling and $q \neq 1$	Article proof	Exact dilation and positivity calculation; scalar part rechecked in Lean.
Herglotz and conformal trace identities	Article proof	Standard planar analysis; outside the Lean scope.
Zero scattered field	Article proof	Exterior total field is the incident field identically; uniqueness of the direct problem cited in the article.

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